EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

Department of Artificial Intelligence & Data Science
Statistical Machine Learning for Data Science (BAD702)
Mini Project Review

Project Title

HOTEL BOOKING CANCELLATION PREDICTION USING MACHINE LEARNING

Presented By:
K S KEERTHI (1EP23AD018)
VINOD KUMAR A(1EP23AD060)

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INTRODUCTION & PROBLEM STATEMENT

- Hotel booking cancellations can cause **revenue loss and inefficient resource management**.
- Traditional systems mainly record bookings but may not predict whether a customer will cancel.
- Machine Learning can identify **patterns in historical booking data**.
- The proposed system predicts whether a booking is likely to be:
  - **Cancelled**
  - **Not Cancelled**
  - The system also provides a **cancellation probability and risk level**

ABSTRACT

- The project develops a **Hotel Booking Cancellation Prediction System**.
- Historical hotel booking data is used to train a machine learning classification model.
- Important booking attributes are processed before prediction.
- A Flask backend connects the trained model with the web application.
- Users enter booking details through a simple frontend.

- The system generates:
  - Cancellation prediction
  - Cancellation probability
  - Risk level

- A dashboard stores and displays previous predictions.

EXISTING SYSTEM AND PROPOSED SYSTEM

Existing System

Traditional hotel booking systems

mainly focus on:

- Booking registration
- Customer information
- Room availability
- Reservation management
- Cancellation recording

Proposed System

Our system uses Machine Learning to predict booking cancellation.

Features:

- User-friendly web interface
- Booking details as input
- Automated data preprocessing
- Machine learning prediction
- Cancellation probability
- Low / Medium / High risk classification
- Dashboard for prediction history
- Visualization using charts

HARDWARE AND SOFTWARE REQUIREMENTS

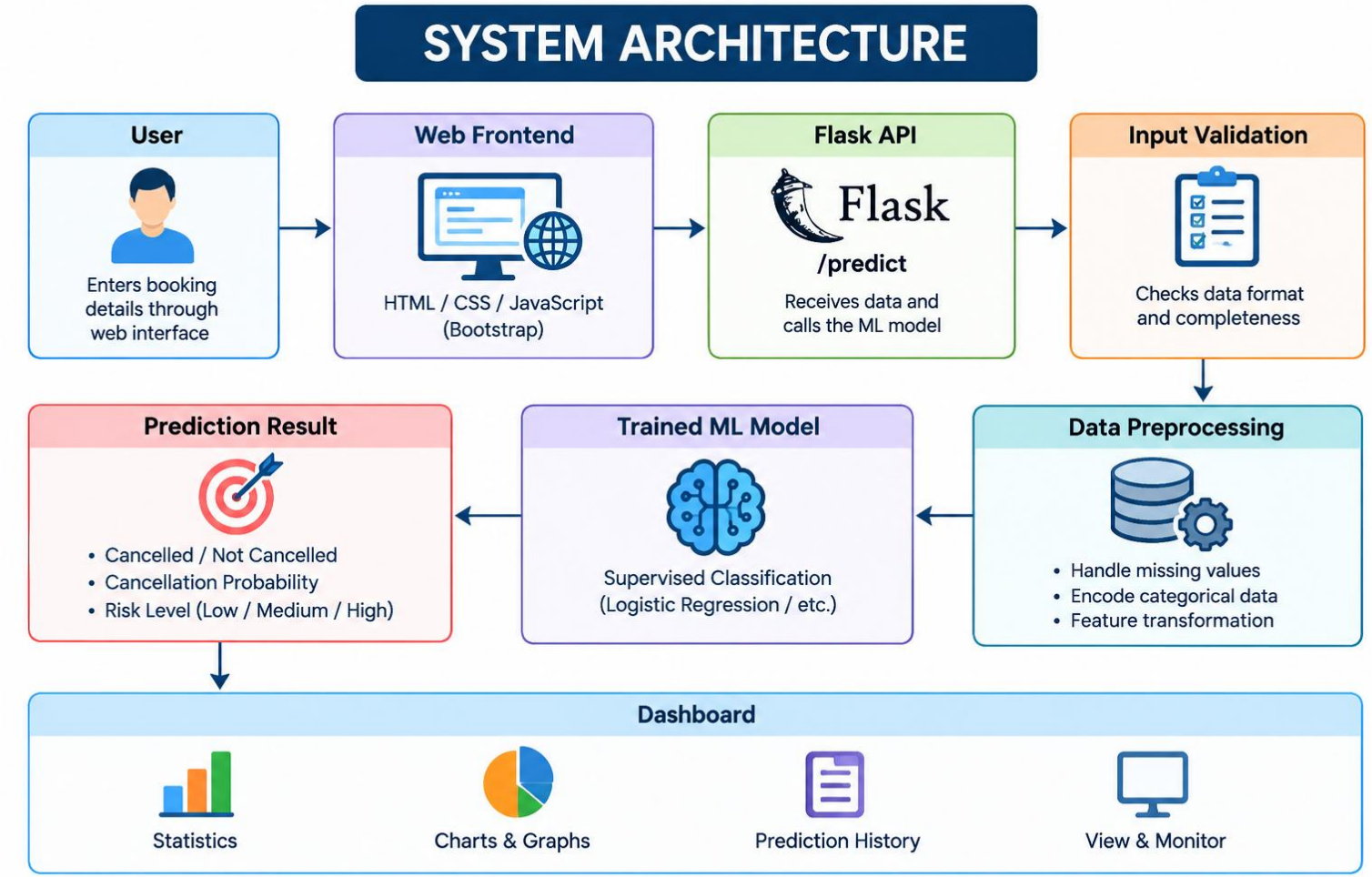
Hardware Requirements

- Processor: Intel Core i3/i5 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 5 GB free space
- Display: Standard monitor
- Input: Keyboard & Mouse

Software Requirements

- OS: Windows / Linux
- Language: Python
- Frontend: HTML, CSS, JavaScript, Bootstrap
- Backend: Flask
- ML/Data: Scikit-learn, Pandas, Joblib
- Visualization: Chart.js
- IDE: VS Code
- Browser: Chrome / Microsoft Edge

SYSTEM ARCHITECHTURE



ALGORITHM AND TECHNOLOGY

Algorithm

- Supervised Machine Learning – Binary Classification
- Collect hotel booking dataset
- Clean and preprocess the data
- Select relevant booking features
- Encode categorical variables
- Split data into training and testing sets
- Train the classification model
- Evaluate the model
- Predict Cancelled / Not Cancelled
- Calculate cancellation probability
- Assign Low / Medium / High Risk

Technologies Used

- Python – Machine Learning & backend
- Pandas – Data processing
- Scikit-learn – ML & preprocessing
- Joblib – Model saving/loading
- Flask – REST API backend
- HTML, CSS, JavaScript – Frontend
- Bootstrap – User interface
- Chart.js – Dashboard visualization

IMPLEMENTATION

- **Data Collection:** Hotel booking dataset is collected and analyzed.
- **Data Preprocessing:** Missing values are handled and categorical data is converted into a suitable format.
- **Model Training:** The processed data is used to train a supervised ML classification model.
- **Model Saving:** Trained model and preprocessing pipeline are saved using **Joblib**.
- **Backend:** Flask API receives booking details and performs prediction.
- **Frontend:** HTML, CSS, JavaScript and Bootstrap provide a simple booking interface.
- **Prediction:** System displays **Cancelled / Not Cancelled**, probability and risk level.
- **Dashboard:** Prediction history and statistics are displayed using **Chart.js**.

REFERENCES & CONCLUSION

References

- **Kaggle** – Hotel Booking Demand Dataset
- **Scikit-learn Documentation** – Machine Learning and preprocessing
- **Pandas Documentation** – Data analysis and data processing
- **Flask Documentation** – Web application and REST API development
- **Bootstrap Documentation** – Frontend interface design
- **Chart.js Documentation** – Data visualization

Conclusion

- Developed a Machine Learning-based Hotel Booking Cancellation Prediction System.
- The system predicts whether a booking is Cancelled or Not Cancelled.
- It provides cancellation probability and risk level.
- Flask connects the trained ML model with the web interface.
- The dashboard helps visualize and monitor prediction results.
- The project demonstrates the practical application of Machine Learning in the hospitality domain.